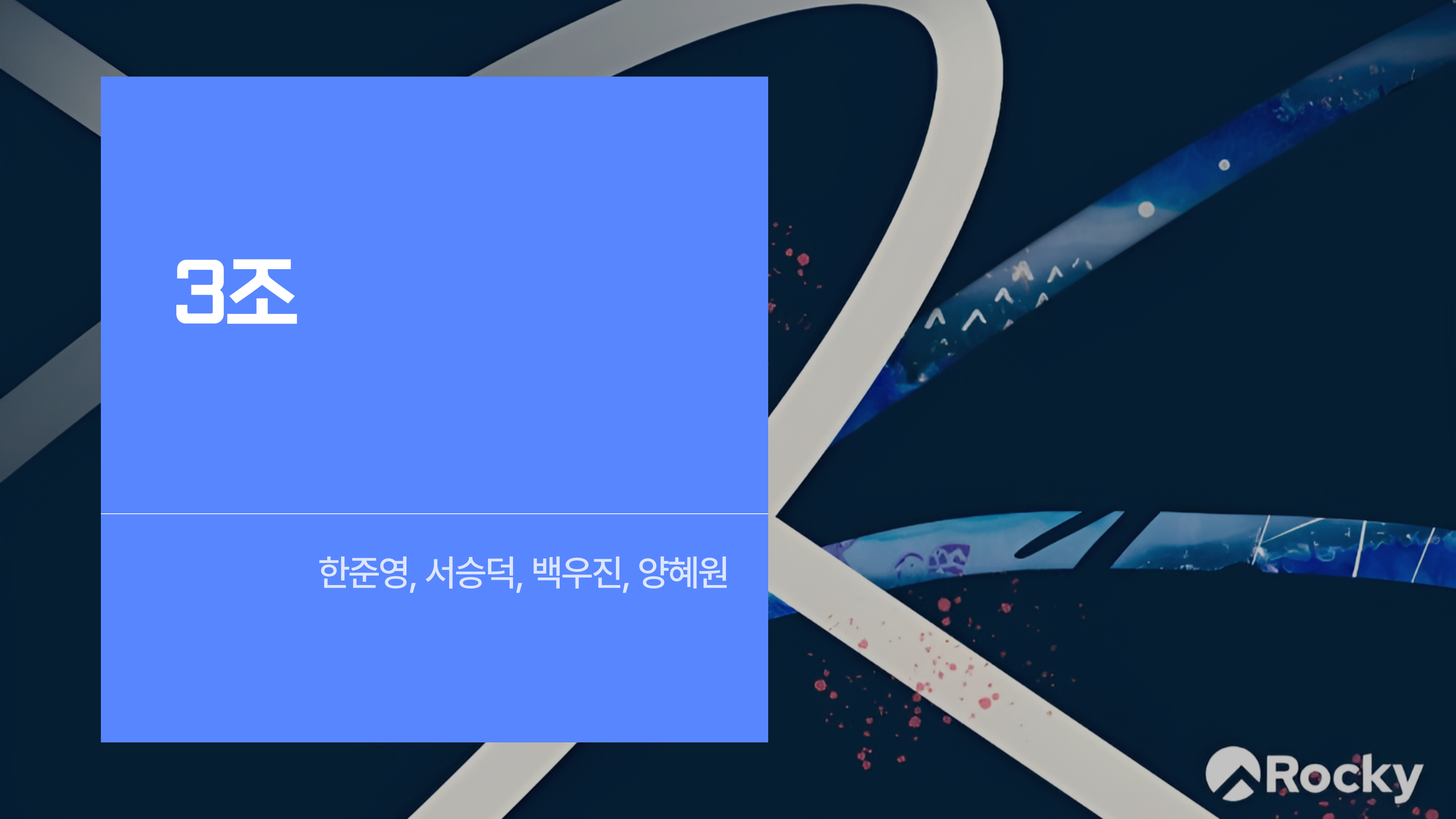
3조

한준영, 서승덕, 백우진, 양혜원



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01

설치

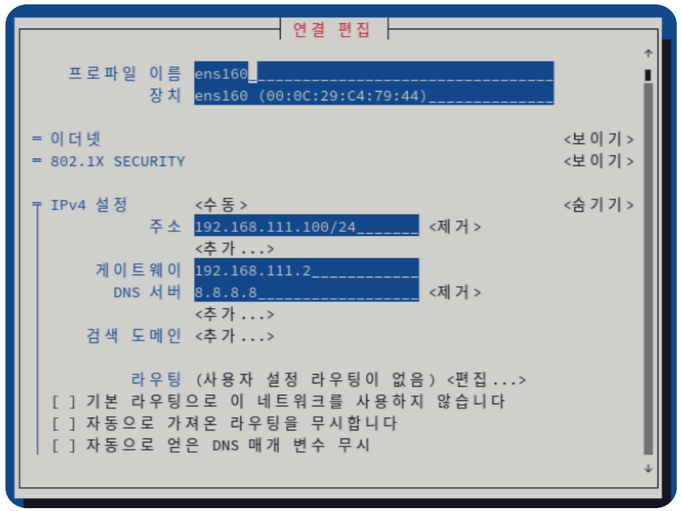
- 주소 설정
- 주소 변경 전 & 변경 후

01

설치

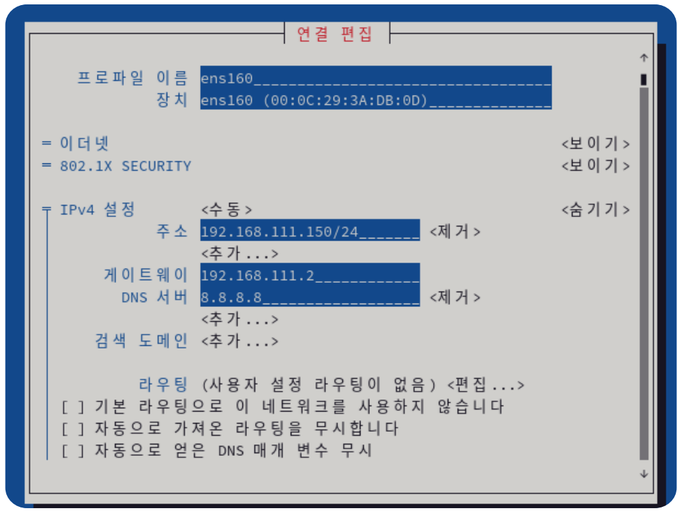
주소 설정

server A



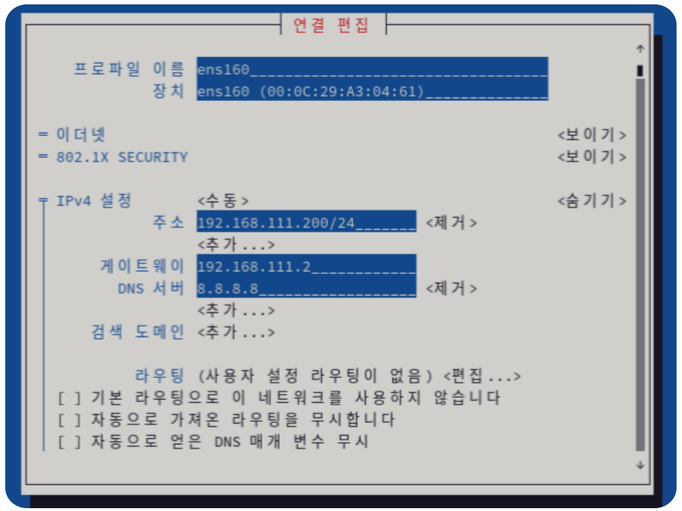
주소 : 192.168.111.100/24
게이트웨어 : 192.168.111.2
DNS 서버 : 8.8.8.8

server B



주소 : 192.168.111.150/24
게이트웨어 : 192.168.111.2
DNS 서버 : 8.8.8.8

server C



주소 : 192.168.111.200/24
게이트웨어 : 192.168.111.2
DNS 서버 : 8.8.8.8

01

설치

주소 변경 전 & 변경 후

server A

```
[rocky@rocky ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.253.130  netmask 255.255.255.0  broadcast 192.168.253.255
        inet6 fe80::20c:29ff:fec4:7944  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:c4:79:44  txqueuelen 1000  (Ethernet)
        RX packets 14536  bytes 21514669 (20.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3906  bytes 230273 (224.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

↓

```
[rocky@rocky ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.111.100  netmask 255.255.255.0  broadcast 192.168.111.255
        inet6 fe80::20c:29ff:fec4:7944  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:c4:79:44  txqueuelen 1000  (Ethernet)
        RX packets 14546  bytes 21516428 (20.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3950  bytes 234228 (228.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

192.168.253.130

↓

192.168.111.100

server B

```
[rocky@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.253.129  netmask 255.255.255.0  broadcast 192.168.253.255
        inet6 fe80::20c:29ff:fe3a:db0d  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:3a:db:0d  txqueuelen 1000  (Ethernet)
        RX packets 14585  bytes 21513104 (20.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4478  bytes 260542 (254.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

↓

```
[rocky@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.111.150  netmask 255.255.255.0  broadcast 192.168.111.255
        inet6 fe80::20c:29ff:fe3a:db0d  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:3a:db:0d  txqueuelen 1000  (Ethernet)
        RX packets 14751  bytes 21537776 (20.5 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4627  bytes 276489 (270.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

192.168.253.129

↓

192.168.111.150

server C

```
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.253.128  netmask 255.255.255.0  broadcast 192.168.253.255
        inet6 fe80::20c:29ff:fea3:461  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a3:04:61  txqueuelen 1000  (Ethernet)
        RX packets 728595  bytes 1094313795 (1.0 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 135020  bytes 7332228 (6.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

↓

```
[rocky@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.111.200  netmask 255.255.255.0  broadcast 192.168.111.255
        inet6 fe80::20c:29ff:fea3:461  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a3:04:61  txqueuelen 1000  (Ethernet)
        RX packets 728876  bytes 1094697026 (1.0 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 135110  bytes 7338696 (6.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

192.168.253.128

↓

192.168.111.200

02

사용자 및 그룹 등록

- 사용자 생성
- 그룹 생성 및 등록

02

사용자 및 그룹 등록

사용자 생성

사용자 생성

```
[root@localhost ~]# useradd sonhm
[root@localhost ~]# useradd leegi
[root@localhost ~]# useradd kimmj
[root@localhost ~]# useradd hwanghc
[root@localhost ~]# useradd hanjy
[root@localhost ~]# useradd seosd
[root@localhost ~]# useradd yanghw
[root@localhost ~]# useradd baeguj
```

생성된 사용자 확인

```
[root@localhost ~]# cat /etc/passwd | tail -8
sonhm:x:1001:1001::/home/sonhm:/bin/bash
leegi:x:1002:1002::/home/leegi:/bin/bash
kimmj:x:1003:1003::/home/kimmj:/bin/bash
hwanghc:x:1004:1004::/home/hwanghc:/bin/bash
hanjy:x:1005:1005::/home/hanjy:/bin/bash
seosd:x:1006:1006::/home/seosd:/bin/bash
yanghw:x:1007:1007::/home/yanghw:/bin/bash
baeguj:x:1008:1008::/home/baeguj:/bin/bash
```

02

사용자 및 그룹 등록

그룹 생성 및 등록

그룹 생성

```
[root@localhost ~]# groupadd eusoccer
[root@localhost ~]# groupadd krsoccer
```

그룹 등록

```
[root@localhost ~]# gpasswd -a sonhm eusoccer
사용자 sonhm을(를) eusoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a leegi eusoccer
사용자 leegi을(를) eusoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a kimmj eusoccer
사용자 kimmj을(를) eusoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a hwanghc eusoccer
사용자 hwanghc을(를) eusoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a hanjy krsoccer
사용자 hanjy을(를) krsoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a seosd krsoccer
사용자 seosd을(를) krsoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a yanghw krsoccer
사용자 yanghw을(를) krsoccer 그룹에 등록 중
[root@localhost ~]# gpasswd -a baeguj krsoccer
사용자 baeguj을(를) krsoccer 그룹에 등록 중
```

그룹에 사용자 등록

그룹 등록 확인

eusoccer 그룹에 포함된 사용자 확인

```
[root@localhost ~]# cat /etc/group | tail -2
eusoccer:x:1009:sonhm,leegi,kimmj,hwanghc
krsoccer:x:1010:hanjy,seosd,yanghw,baeguj
```

krsoccer 그룹에 포함된 사용자 확인

03

디스크 추가 후 LVM 설정

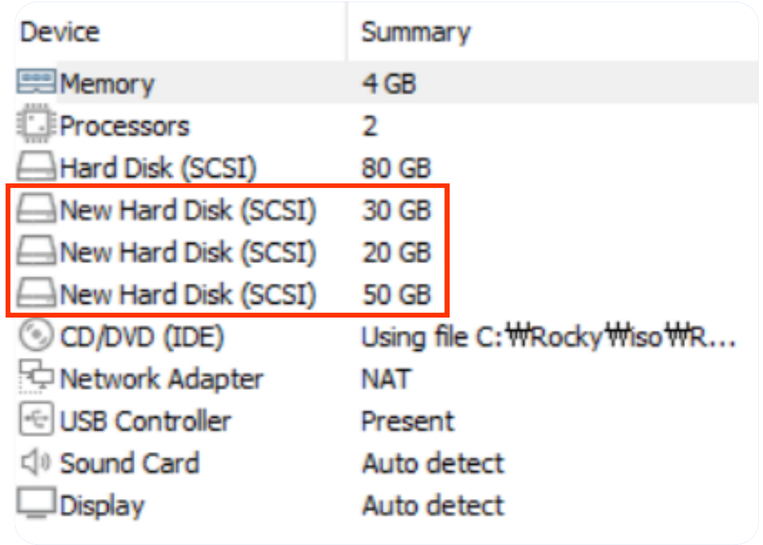
- 디스크 추가
- 파티션 생성
- LVM 설정

03

디스크 추가 후 LVM 설정

디스크 추가

하드 디스크 추가



디스크 정보 확인

```
[root@rocky ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   80G  0 disk
├─sda1   8:1    0    8G  0 part [SWAP]
└─sda2   8:2    0   72G  0 part /
sdb      8:16   0   50G  0 disk
sdc      8:32   0   30G  0 disk
sdd      8:48   0   20G  0 disk
sr0     11:0    1 1024M  0 rom
```

03

디스크 추가 후 LVM 설정

파티션 생성

```
[root@rocky ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xb15e437d.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1): 1
First sector (2048-104857599, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-104857599, default 104857599):

Created a new partition 1 of type 'Linux' and of size 50 GiB.
```

새로운 파티션 생성

파티션 종류 설정, 기본값 : p

파티션 번호 선택

파일 시스템 유형 선택

```
Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p
Disk /dev/sdb: 50 GiB, 53687091200 bytes, 104857600 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb15e437d

Device     Boot Start       End   Sectors Size Id Type
/dev/sdb1        2048 104857599 104855552  50G 8e Linux LVM
```

파일 시스템 유형

설정 내용 확인

설정 내용 확인

03

디스크 추가 후 LVM 설정

LVM 설정

PV 구성

*PV : 실제 하드디스크 파티션

```
[root@rocky ~]# pvcreate /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
[root@rocky ~]# pvcreate /dev/sdc1
  Physical volume "/dev/sdc1" successfully created.
[root@rocky ~]# pvcreate /dev/sdd1
  Physical volume "/dev/sdd1" successfully created.
[root@rocky ~]# pvscan
  PV /dev/sdb1         lvm2 [<50.00 GiB]
  PV /dev/sdc1         lvm2 [<30.00 GiB]
  PV /dev/sdd1         lvm2 [<20.00 GiB]
  Total: 3 [<100.00 GiB] / in use: 0 [0   ] / in no VG: 3 [<100.00 GiB]
```

PV생성

PV 설정 내용 확인

VG 구성

*VG : 여러 개의 PV를 그룹으로 묶은 것

```
[root@rocky ~]# vgcreate DATA /dev/sdb1 /dev/sdc1 /dev/sdd1
  Volume group "DATA" successfully created
[root@rocky ~]# vgdisplay
  --- Volume group ---
  VG Name               DATA
  System ID
  Format                lvm2
  Metadata Areas        3
  Metadata Sequence No  1
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                0
  Open LV               0
  Max PV                0
  Cur PV                3
  Act PV                3
  VG Size               <99.99 GiB
  PE Size               4.00 MiB
  Total PE              25597
  Alloc PE / Size       0 / 0
  Free  PE / Size       25597 / <99.99 GiB
  VG UUID               Kw2kRE-Odzk-XSs9-Wluk-Vt7C-wN8h-hpnWuM
```

VG 설정 내용 확인

VG 생성

LV 구성

*LV : VG를 다시 적절한 크기의 파티션으로 나눌 때

```
[root@rocky ~]# lvcreate --size 40G --name VIDEO DATA
  Logical volume "VIDEO" created.
[root@rocky ~]# lvcreate --extents 100%FREE --name AUDIO DATA
  Logical volume "AUDIO" created.
[root@rocky ~]# lvscan
  ACTIVE            '/dev/DATA/VIDEO' [40.00 GiB] inherit
  ACTIVE            '/dev/DATA/AUDIO' [<59.99 GiB] inherit
```

LV 생성

나머지 용량 AUDIO에 모두 할당

LV 생성 내용 확인

03

디스크 추가 후 LVM 설정

LVM 설정

LVM 삽입 확인

```
sdb             8:16   0   50G  0 disk
└─sdb1          8:17   0   50G  0 part
  ├─DATA-VIDEO 253:0    0   40G  0 lvm
  └─DATA-AUDIO 253:1    0   60G  0 lvm
sdc             8:32   0   30G  0 disk
└─sdc1          8:33   0   30G  0 part
  └─DATA-AUDIO 253:1    0   60G  0 lvm
sdd             8:48   0   20G  0 disk
└─sdd1          8:49   0   20G  0 part
  └─DATA-AUDIO 253:1    0   60G  0 lvm
```

DATA-VIDEO 삽입

DATA-AUDIO 삽입

파일 시스템 생성

```
[root@rocky ~]# mkfs.ext4 /dev/DATA/VIDEO
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 10485760 4k blocks and 2621440 inodes
Filesystem UUID: 9dc3d711-dc44-4a91-bf74-6a4845b4b169
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624

Allocating group tables: done
Writing inode tables: done
Creating journal (65536 blocks): done
Writing superblocks and filesystem accounting information: done

[root@rocky ~]# mkfs.ext4 /dev/DATA/AUDIO
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 15725568 4k blocks and 3932160 inodes
Filesystem UUID: 12de9e7a-b7a1-43e1-8bb9-bacb744f7b9a
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000, 7962624, 11239424

Allocating group tables: done
Writing inode tables: done
Creating journal (65536 blocks): done
Writing superblocks and filesystem accounting information: done
```

/dev/DATA/VIDEO 파일 시스템 생성

/dev/DATA/AUDIO 파일 시스템 생성

03

디스크 추가 후 LVM 설정

LVM 설정

마운트

*장치명
*마운트 포인트

```
[root@rocky ~]# mkdir /lvm1 /lvm2
[root@rocky ~]# mount /dev/DATA/VIDEO /lvm1
[root@rocky ~]# mount /dev/DATA/AUDIO /lvm2
```

/dev/DATA/VIDEO를 /lvm1에 마운트
/dev/DATA/AUDIO를 /lvm2에 마운트

내용 수정

*/etc/fstab 파일

```
/dev/DATA/VIDEO         /lvm1   ext4    defaults        0       0
/dev/DATA/AUDIO         /lvm2   ext4    defaults        0       0
```

/dev/DATA/VIDEO의 내용 수정
/dev/DATA/AUDIO의 내용 수정

삽입 확인

/lvm1 삽입 확인

```
sdb                 8:16   0   50G  0 disk
└─sdb1              8:17   0   50G  0 part
  ├─DATA-VIDEO 253:0    0   40G  0 lvm  /lvm1
  └─DATA-AUDIO 253:1    0   60G  0 lvm  /lvm2
sdc                 8:32   0   30G  0 disk
└─sdc1              8:33   0   30G  0 part
  └─DATA-AUDIO 253:1    0   60G  0 lvm  /lvm2
sdd                 8:48   0   20G  0 disk
└─sdd1              8:49   0   20G  0 part
  └─DATA-AUDIO 253:1    0   60G  0 lvm  /lvm2
sr0                11:0    1 1024M  0 rom
```

/lvm2 삽입 확인

04

디스크 쿼터 설정

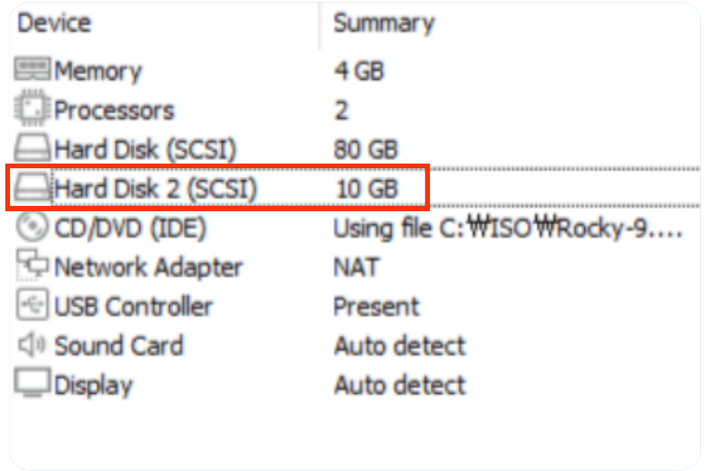
- 디스크 추가
- 디스크 설정
- 쿼터 설정

04

디스크 쿼터 설정

하드 디스크 추가

하드 디스크 추가



디스크 정보 확인

```
[root@server ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   80G  0 disk
├─sda1   8:1    0    8G  0 part [SWAP]
└─sda2   8:2    0   72G  0 part /
sdb      8:16   0   10G  0 disk
└─sdb1   8:17   0   10G  0 part
```

04

디스크 쿼터 설정

디스크 설정

마운트

/etc/fstab 파일

```
/dev/sdb1       /renew  ext4    defaults,usrjquota=aquota.user,jqfmt=vfsv0      0       0
```

→ 자동 마운트 설정

마운트 재시작 및 확인

```
[root@localhost ~]# mount --options remount /userHome
[root@localhost ~]# mount | grep /userHome
/dev/sdb1 on /userHome type ext4 (rw,relatime,seclabel,jqfmt=vfsv0,usrjquota=aquota.user)
```

→ 마운트 재시작

→ 마운트 확인

04

디스크 쿼터 설정

쿼터 설정

쿼터 DB 생성

쿼터 종료

```
[root@localhost ~]# quotaoff -avug
quotaoff: Your kernel probably supports ext4 quota feature but you are using external quota files. Please switch your filesystem to use ext4 quota
 feature as external quota files on ext4 are deprecated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <d
evice>'.
/dev/sdb1 [/userHome]: user quotas turned off
[root@localhost ~]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota files. Please switch your filesystem to use ext4 quo
ta feature as external quota files on ext4 are deprecated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota
<device>'.
[root@localhost ~]# rm -rf aquota.*
[root@localhost ~]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota files. Please switch your filesystem to use ext4 quo
ta feature as external quota files on ext4 are deprecated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota
<device>'.
[root@localhost ~]# touch aquota.user aquota.group
[root@localhost ~]# chmod 600 aquota.*
[root@localhost ~]# quotacheck -augmn
quotacheck: Your kernel probably supports ext4 quota feature but you are using external quota files. Please switch your filesystem to use ext4 quo
ta feature as external quota files on ext4 are deprecated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota
<device>'.
[root@localhost ~]# quotaon -avug
quotaon: Your kernel probably supports ext4 quota feature but you are using external quota files. Please switch your filesystem to use ext4 quota
feature as external quota files on ext4 are deprecated. You can enable the feature by unmounting the file system and running 'tune2fs -O quota <de
vice>'.
/dev/sdb1 [/userHome]: user quotas turned on
```

쿼터 관련 사항 체크

쿼터 관련 파일 삭제

쿼터 관련 파일 생성

소유자 외 접근 금지 설정

쿼터 시작

04

디스크 쿼터 설정

쿼터 설정

사용자 공간 할당

```
[root@localhost userHome]# edquota -u aespa
[root@localhost userHome]# edquota -u IVE
[root@localhost userHome]# edquota -u NewJeans
```

사용자 할당량 편집

사용 한도 제한 확인

```
[root@localhost userHome]# repquota /userHome
*** Report for user quotas on device /dev/sdb1
Block grace time: 7days; Inode grace time: 7days
                        Block limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --       20       0       0              3     0     0
aespa     --       28  716800 1048576              7     0     0
IVE       --       28  716800 1048576              7     0     0
NewJeans  --       28  716800 1048576              7     0     0
```

사용자별 현재 할당량 확인

각각 700M , 1G

05

서버 구성

- SSH
- XRDP
- DNS(Web, FTP)
- MariaDB
- Samba
- DHCP
- Mail

05

서버 구성

SSH

설치 및 확인

```
[root@localhost ~]# rpm -qa openssh-server
openssh-server-8.7p1-46.el9.rocky.0.1.x86_64
```

SSH 설치 여부 확인

설치 완료

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client ssh
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

활성화 확인

```
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-01-12 15:27:14 KST; 23h ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 876 (sshd)
      Tasks: 1 (limit: 22782)
     Memory: 3.1M (peak: 3.3M)
        CPU: 18ms
     CGroup: /system.slice/sshd.service
             └─876 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

1월 12 15:27:14 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
1월 12 15:27:14 localhost.localdomain sshd[876]: Server listening on 0.0.0.0 port 22.
1월 12 15:27:14 localhost.localdomain sshd[876]: Server listening on :: port 22.
1월 12 15:27:14 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

SSH 활성화 상태 확인

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서버 구성

XRDP

설치 및 확인

```
[root@localhost ~]# rpm -qa epel-release
epel-release-9-10.el9.noarch
```

설치 완료

XRDP 설치 여부 확인

활성화 확인

```
[root@localhost ~]# systemctl status xrdp
● xrdp.service - xrdp daemon
     Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-01-13 15:04:30 KST; 1min 7s ago
       Docs: man:xrdp(8)
             man:xrdp.ini(5)
   Main PID: 30911 (xrdp)
      Tasks: 1 (limit: 22782)
     Memory: 1.6M (peak: 1.9M)
        CPU: 17ms
     CGroup: /system.slice/xrdp.service
             └─30911 /usr/sbin/xrdp --nodaemon
```

XRDP 활성화 확인

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-port=3389/tcp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-ports
3389/tcp
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

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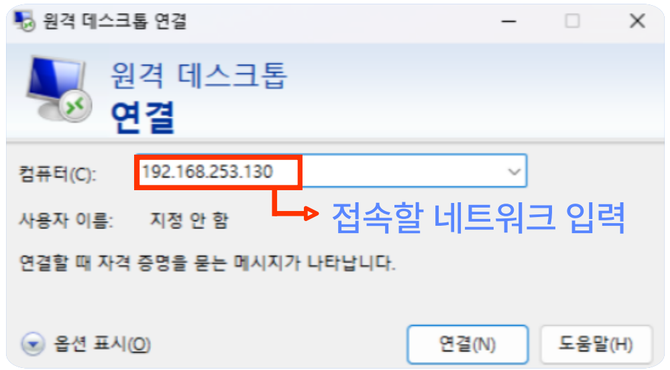
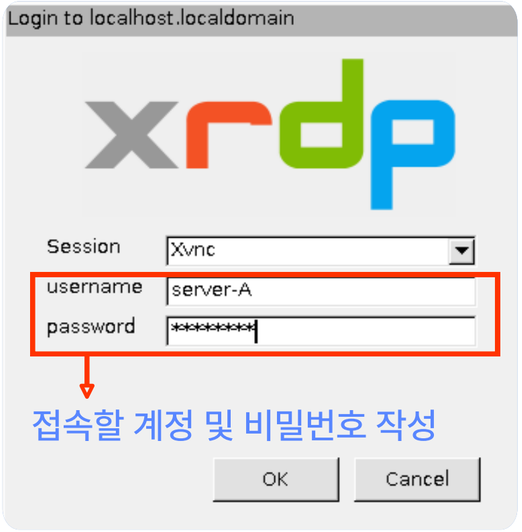
서버 구성

XRDP

서버 테스트





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서버 구성

DNS

설치 및 확인

```
[root@localhost ~]# dnf -y install bind bind-chroot
```
→ 패키지 설치

```
[root@localhost ~]# rpm -qa bind bind-chroot
bind-9.16.23-34.el9_7.1.x86_64
bind-chroot-9.16.23-34.el9_7.1.x86_64
```
→ 패키지 설치 확인

환경 설정

/etc/named.conf 파일

```
10 options {
11         listen-on port 53 { 127.0.0.1; };
12         listen-on-v6 port 53 { ::1; };
13         directory       "/var/named";
14         dump-file       "/var/named/data/cache_dump.db";
15         statistics-file "/var/named/data/named_stats.txt";
16         memstatistics-file "/var/named/data/named_mem_stats.txt";
17         secroots-file   "/var/named/data/named.secroots";
18         recursing-file  "/var/named/data/named.recursing";
19         allow-query     { localhost; };
```

↓ DNS 설정 파일 수정

```
10 options {
11         listen-on port 53 { any; };
12         listen-on-v6 port 53 { none; };
13         directory       "/var/named";
14         dump-file       "/var/named/data/cache_dump.db";
15         statistics-file "/var/named/data/named_stats.txt";
16         memstatistics-file "/var/named/data/named_mem_stats.txt";
17         secroots-file   "/var/named/data/named.secroots";
18         recursing-file  "/var/named/data/named.recursing";
19         allow-query     { any; };
```

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서버 구성

DNS

서비스 시작 및 활성화 확인

```
[root@localhost ~]# systemctl restart named
[root@localhost ~]# systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
[root@localhost ~]# systemctl status named
● named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-01-13 16:02:53 KST; 14s ago
   Main PID: 38036 (named)
      Tasks: 8 (limit: 22782)
     Memory: 26.8M (peak: 27.5M)
        CPU: 80ms
     CGroup: /system.slice/named.service
             └─38036 /usr/sbin/named -u named -c /etc/named.conf
```

서비스 재시작

서비스 활성화

DNS 활성화 확인

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-service=dns
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns ssh
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

서버 동작 확인

```
[root@localhost ~]# nslookup
> server 192.168.111.100
Default server: 192.168.111.100
Address: 192.168.111.100#53
```

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서버 구성

DNS

환경 설정

```
[root@localhost named]# touch secteam.com.db
[root@localhost named]# ls
chroot  data  dynamic  named.ca  named.empty  named.localhost  named.loopback  secteam.com.db  slaves
```

정방향 영역 파일 생성

/etc/named.conf 파일

```
60 zone "secteam.com" IN {
61         type master;
62         file "secteam.com.db";
63         allow-update { none; };
```

도메인 설정

```
[root@localhost named]# named-checkconf
[root@localhost named]#
```

이상 여부 확인

정방향 영역 파일 수정

```
$TTL    3H
@       SOA     @       root.   ( 2 1D 1H 1W 1H )
        IN      NS      @
        IN      A       192.168.253.130

server-A        IN      A       192.168.253.130
server-B        IN      A       192.168.253.129

www     IN      CNAME   server-A
ftp     IN      CNAME   server-B
```

/var/named/secteam.com.db 파일

```
[root@localhost named]# named-checkzone secteam.com secteam.com.db
zone secteam.com/IN: loaded serial 2
OK
```

정방향 영역 파일 문법 확인

05

서버 구성

Web

설치 및 확인

```
[root@rocky ~]# dnf install -y httpd
[root@rocky ~]# rpm -qa httpd
httpd-2.4.62-7.el9_7.3.x86_64
```

패키지 설치

패키지 설치 확인

서비스 시작 및 활성화 확인

서버 동작 확인

```
[root@rocky ~]# systemctl restart httpd
[root@rocky ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@rocky ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-01-13 17:02:03 KST; 19s ago
       Docs: man:httpd.service(8)
   Main PID: 32859 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 22782)
     Memory: 14.0M (peak: 14.3M)
        CPU: 117ms
```

서비스 재시작

서비스 활성화

httpd 활성화 확인

방화벽 설정

방화벽 허용

```
[root@rocky ~]# firewall-cmd --permanent --add-service=http
success
[root@rocky ~]# firewall-cmd --reload
success
[root@rocky ~]# firewall-cmd --list-service
cockpit dhcpv6-client ftp http ssh
```

방화벽 재가동

방화벽 가동 확인

웹 페이지 설정 및 테스트

```
[root@rocky ~]# cd /var/www/html
[root@rocky html]# ls
[root@rocky html]# vi /var/www/html/index.html
<h1> this is Linux Web Server </h1>
```

http://www.thisislinux.com

MJC cloud service

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서버 구성

FTP

설치 및 확인

```
[root@rocky ~]# dnf -y install vsftpd
[root@rocky ~]# rpm -qa vsftpd
vsftpd-3.0.5-6.el9.x86_64
```

패키지 설치

패키지 설치 여부 확인

환경 설정

```
anonymous_enable=YES
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
local_umask=022
```

방화벽 설정

```
[root@rocky ~]# firewall-cmd --permanent --add-service=ftp
success
[root@rocky ~]# firewall-cmd --reload
success
[root@rocky ~]# firewall-cmd --list-service
cockpit dhcpv6-client ftp ssh
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

서비스 시작 및 활성화 확인

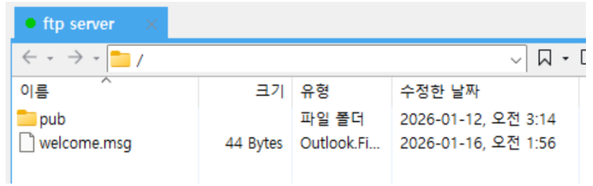
```
[root@rocky ~]# systemctl restart vsftpd
[root@rocky ~]# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.
[root@rocky ~]# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: d
     Active: active (running) since Tue 2026-01-13 16:38:41 KST; 10s ago
   Main PID: 32115 (vsftpd)
      Tasks: 1 (limit: 22782)
     Memory: 860.0K (peak: 1.2M)
        CPU: 3ms
     CGroup: /system.slice/vsftpd.service
             └─32115 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf
```

서비스 재시작

서비스 활성화

vsftpd 활성화 확인

서버 동작 확인



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서버 구성

NFS

설치 및 확인

```
[root@localhost ~]# rpm -qa nfs-utils
nfs-utils-2.5.4-38.el9.x86_64
```

설치 완료

패키지 설치 여부 확인

권한 설정

```
root@localhost:
/share          *(rw,sync)
```

```
[root@localhost ~]# chmod 707 share
[root@localhost ~]# ls -l | grep share
drwx---rwx. 2 root root    6  1월  14 17:01 share
```

/share 디렉터리에
해당 IP 주소 컴퓨터 접근 허용 및 권한 부여

서비스 시작 및 활성화 확인

```
[root@localhost ~]# systemctl restart nfs-server
[root@localhost ~]# systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server
ervice.
[root@localhost ~]# systemctl status nfs-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: disabled)
    Drop-In: /run/systemd/generator/nfs-server.service.d
             └─order-with-mounts.conf
     Active: active (exited) since Wed 2026-01-14 17:10:09 KST; 26s ago
       Docs: man:rpc.nfsd(8)
             man:exportfs(8)
   Main PID: 41013 (code=exited, status=0/SUCCESS)
        CPU: 45ms

 1월 14 17:10:08 localhost.localdomain systemd[1]: Starting NFS server and services...
 1월 14 17:10:09 localhost.localdomain systemd[1]: Finished NFS server and services.
```

서비스 재시작
서비스 활성화
활성화 확인

```
[root@localhost ~]# exportfs -v
/share          <world>(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
```

NFS 서비스 가동 유무 확인

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서버 구성

NFS

방화벽 설정

방화벽 허용

```
[root@localhost ~]# firewall-cmd --permanent --add-service=nfs
success
[root@localhost ~]# firewall-cmd --permanent --add-service=mountd
success
[root@localhost ~]# firewall-cmd --permanent --add-service=rpc-bind
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns mountd nfs rpc-bind ssh
```

방화벽 재가동

방화벽 가동 확인

NFS 설치 및 확인

```
[root@rocky ~]# rpm -qa nfs-utils
nfs-utils-2.5.4-38.el9.x86_64
```

다른 서버 패키지 설치 여부 확인

설치 완료

마운트

```
[root@rocky ~]# showmount -e 192.168.253.130
Export list for 192.168.253.130:
/share *
```

NFS 서버의 공유 디렉터리 확인

```
vmlinuz-0-rescue-9f8673e8b5a74c9d829f2913339e28e5
vmlinuz-5.14.0-611.5.1.el9_7.x86_64
```

NFS 서버 공유 디렉터리를
클라이언트 쪽에서 생성한 디렉터리와 마운트

05

서버 구성

Samba

설치 및 확인

```
[root@localhost named]# rpm -qa samba
samba-4.22.4-6.el9.x86_64
```

패키지 설치 확인

설치 완료

마운트

공유 디렉터리 생성 후 마운트

```
[root@localhost ~]# mount -t cifs //10.11.82.84/share /smshare
Password for root@//10.11.82.84/share:
```

환경 설정

```
workgroup = USERG
unix charset = UT
map to guest = Ba
```

samba 파일 수정

/etc/samba/smb.conf 파일

```
[root@localhost ~]# testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

Press enter to see a dump of your service definitions
```

설정 파일 오류 확인

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서버 구성

Samba

서비스 시작 및 활성화 확인

```
[root@localhost ~]# systemctl restart smb nmb        → 서비스 재시작
[root@localhost ~]# systemctl enable smb nmb         → 서비스 활성화
Created symlink /etc/systemd/system/multi-user.target.wants/smb.service →
ib/systemd/system/smb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/nmb.service →
ib/systemd/system/nmb.service.
[root@localhost ~]# systemctl status smb nmb         → 활성화 확인
● smb.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smb.service; enabled; preset:
     Active: active (running) since Thu 2026-01-15 16:14:25 KST; 15s ago
```

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-service=samba-client   → 방화벽 허용
success
[root@localhost ~]# firewall-cmd --permanent --add-service=samba
Warning: ALREADY_ENABLED: samba
success
[root@localhost ~]# firewall-cmd --reload            → 방화벽 재가동
success
[root@localhost ~]# firewall-cmd --list-services     → 방화벽 가동 확인
cockpit dhcp dhcpv6-client dns mountd nfs rpc-bind samba samba-client ssh
```

SELinux 설정

```
setsebool P samba_enable_home_dirs on
chcon -R -t samba_share_t /smbshare
```

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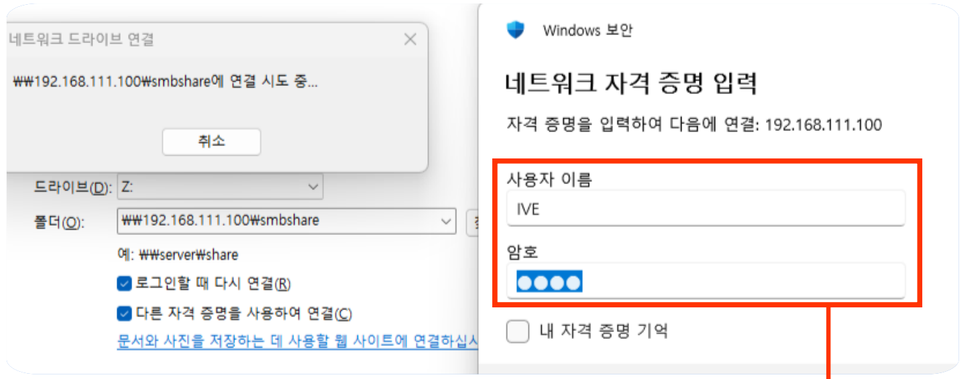
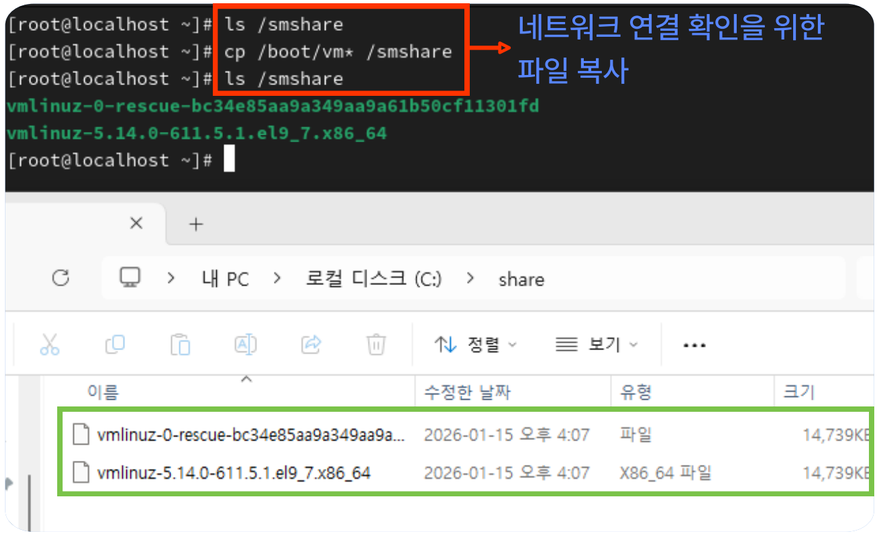
서버 구성

Samba

서버 동작 확인



네트워크 자격 증명 선택 후
계정 이름 및 비밀번호 입력



네트워크 연결 확인을 위한
파일 복사

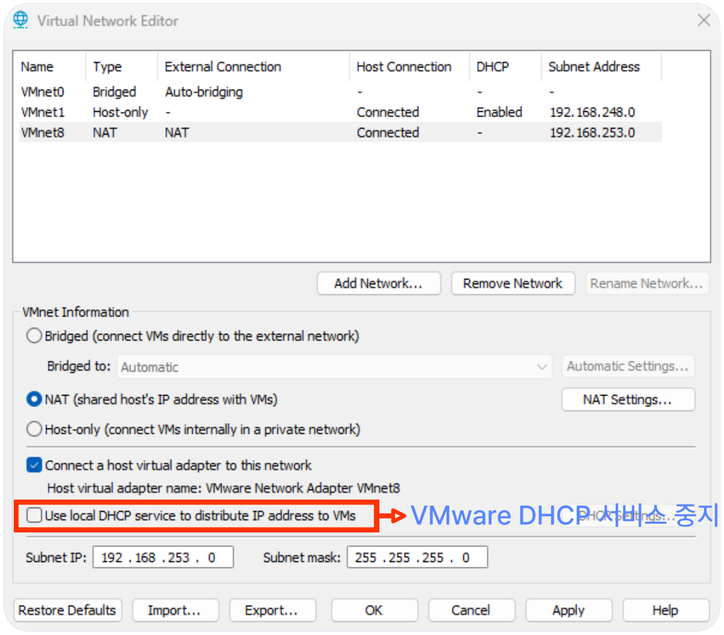
복사한 파일 확인

05

서버 구성

DHCP

DHCP 서비스 중지




```
[root@localhost ~]# ifconfig ens160
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet6 fe80::20c:29ff:fe83:a1c3  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:83:a1:c3  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 41  bytes 6592 (6.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

DHCP 서비스 중지 후 주소 확인

05

서버 구성

DHCP

설치 및 확인

```
[root@localhost ~]# rpm -qa dhcp-client
[root@localhost ~]# dnf -y install dhcp-client
```

패키지 설치확인

설치 완료

환경 설정

/etc/dhcp/dhcpd.conf 파일

```
ddnsupdate-style interim;
dubnet 192.168.111.0 netmask 255.255.255.0 {
        option routers 192.168.111.2;
        option subnet-mask 255.255.255.0;
        range dynamic-bootp 192.168.111.55 192.168.111.99;
        option domain-name-servers 8.8.8.8
        default-lease-time 10000;
        max-lease-time 50000;
```

dhcpd.conf 파일 수정

서비스 시작 및 활성화 확인

```
[root@localhost ~]# systemctl restart dhcpd
[root@localhost ~]# systemctl enable dhcpd
[root@localhost ~]# systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset:
     Active: active (running) since Thu 2026-01-15 15:50:17 KST; 16s ago
```

서비스 재시작

서비스 활성화

활성화 확인

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서버 구성

DHCP

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-service=dhcp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns mountd nfs rpc-bind samba ssh
[root@localhost ~]#
```

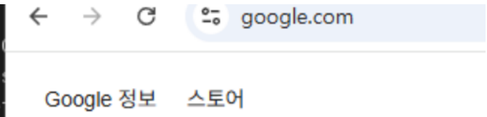
방화벽 허용

방화벽 재가동

방화벽 가동 확인

서버 동작 확인

```
[root@localhost ~]# ifconfig ens160
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 150
        inet 192.168.111.55  netmask 255.255.255.0  broadca
        inet6 fe80::20c:29ff:fec4:7944  prefixlen 64  scope
        ether 00:0c:29:c4:79:44  txqueuelen 1000  (Ethernet
        RX packets 487  bytes 42456 (41.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 346  bytes 35354 (34.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  colli
```



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서버 구성

Mail

설치 및 확인

```
[root@localhost ~]# rpm -qa sendmail
sendmail-8.16.1-11.el9.x86_64
```

패키지 설치확인

설치 완료

환경 설정

```
[root@localhost ~]# hostnamectl set-hostname mail.host.com
[root@localhost ~]# hostname
mail.host.com
```

호스트 이름 수정

설정 파일 수정

```
192.168.111.100 mail.host.com
```

/etc/hosts

```
mail.host.com
```

/etc/sysconfig/network

```
HOSTNAME = mail.host.com
```

/etc/mail/local-host-names

*/etc/dhcp/dhcpd.conf 파일

네임서버 설정 파일 수정 및 추가

```
10 options {
11         listen-on port 53 { 127.0.0.1; };
12         listen-on-v6 port 53 { ::1; };
13         directory       "/var/named";
14         dump-file       "/var/named/data/cache_dump.db";
15         statistics-file "/var/named/data/named_stats.txt";
16         memstatistics-file "/var/named/data/named_mem_stats.txt";
17         secroots-file   "/var/named/data/named.secroots";
18         recursing-file  "/var/named/data/named.recursing";
19         allow-query     { localhost; };
```

```
60 zone "host.com" IN {
61         type master;
62         file "host.cm.db";
63         allow-update { none; };
64         };
```

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서버 구성

Mail

환경 설정

```
root@localhost ~]# named-checkconf
root@localhost ~]#
```

파일 문법 오류 검사

```
root@localhost named]# touch host.com.db
root@localhost named]# ls
dynamic      named.ca      named.localhost  secteam.com.db
host.com.db  named.empty   named.loopback   slaves
```

파일 생성

```
$TTL    3H
@       SOA     @       root.   ( 2       1D      1H      1W      1H )
        IN      NS      @
        IN      A       192.168.111.100
        IN      MX      10      main.host.com.

mail    IN      A       192.168.111.100
```

host.com.db에 내용 추가

```
268 O DaemonPortOptions=Port=smtp, Name=MTA
 85 Cwhost.com
```

/etc/mail/sendmail.cf 내용 수정

```
host.com          RELAY
192.168.111       RELAY
~
```

/etc/mail/access 내용 추가

```
protocols = imap pop3 lmtp submission

# A comma separated list of IPs or hosts wher
# "*" listens in all IPv4 interfaces, "::" li
# If you want to specify non-default ports or
# edit conf.d/master.conf.
listen = *, ::

# Base directory where to store runtime data.
base_dir = /var/run/dovecot/
```

devecot.conf 파일 수정

```
ssl = yes
```

10-ssl.conf 파일 내용 수정

```
mail_location = mbox:~/mail:INBOX=/var/mail/%u
mail_access_groups = mail
lock_method = fcntl
```

10-mail.conf 파일 내용 수정

access 파일 설정 내용 적용

```
[root@localhost ~]# makemap hash /etc/mail/access < /etc/mail/acc
```

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서버 구성

Mail

서비스 시작 및 활성화 확인

```
root@localhost ~]# systemctl restart sendmail
[root@localhost ~]# systemctl restart dovecot
[root@localhost ~]# systemctl enable sendmail
[root@localhost ~]# systemctl enable dovecot
[root@localhost ~]# systemctl status sendmai
Unit sendmai.service could not be found.
[root@localhost ~]# systemctl status sendmail
● sendmail.service - Sendmail Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/sendmail.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-01-15 16:57:21 KST; 25s ago
   Main PID: 6281 (sendmail)
      Tasks: 1 (limit: 22782)
     Memory: 3.8M (peak: 4.1M)
        CPU: 39ms
     CGroup: /system.slice/sendmail.service
             └─6281 "sendmail: accepting connections"

 1월 15 16:57:21 mail.host.com systemd[1]: Starting Sendmail Mail Transport Agent...
 1월 15 16:57:21 mail.host.com sendmail[6281]: starting daemon (8.16.1): SMTP+queueing@01:>
 1월 15 16:57:21 mail.host.com systemd[1]: sendmail.service: Can't open PID file /run/send>
 1월 15 16:57:21 mail.host.com systemd[1]: Started Sendmail Mail Transport Agent.
[root@localhost ~]# systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-01-15 16:57:28 KST; 22s ago
```

서비스 재시작

서비스 활성화

활성화 확인

활성화 확인

방화벽 설정

```
[root@localhost named]# firewall-cmd --permanent --add-service=smtp
success
[root@localhost named]# firewall-cmd --permanent --add-service=pop3
success
[root@localhost named]# firewall-cmd --permanent --add-service=imap
success
[root@localhost named]# firewall-cmd --reload
success
[root@localhost named]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns imap mountd nfs pop3 rpc-bind samba samba-client
smtp ssh
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

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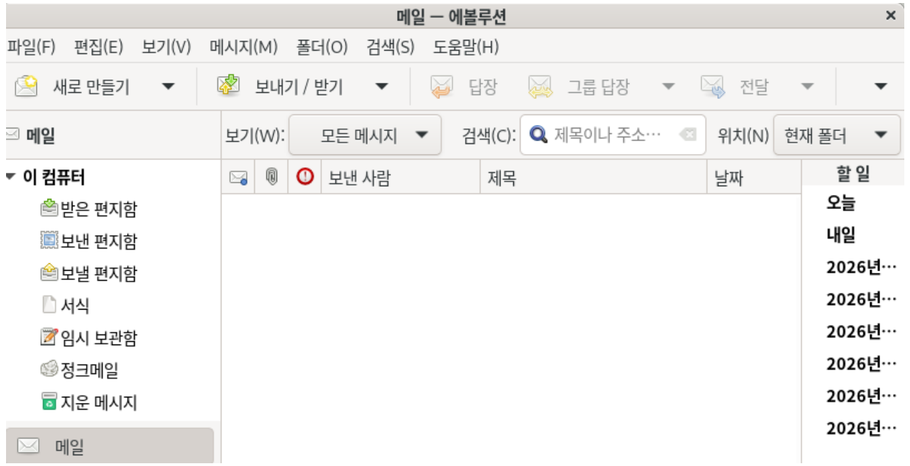
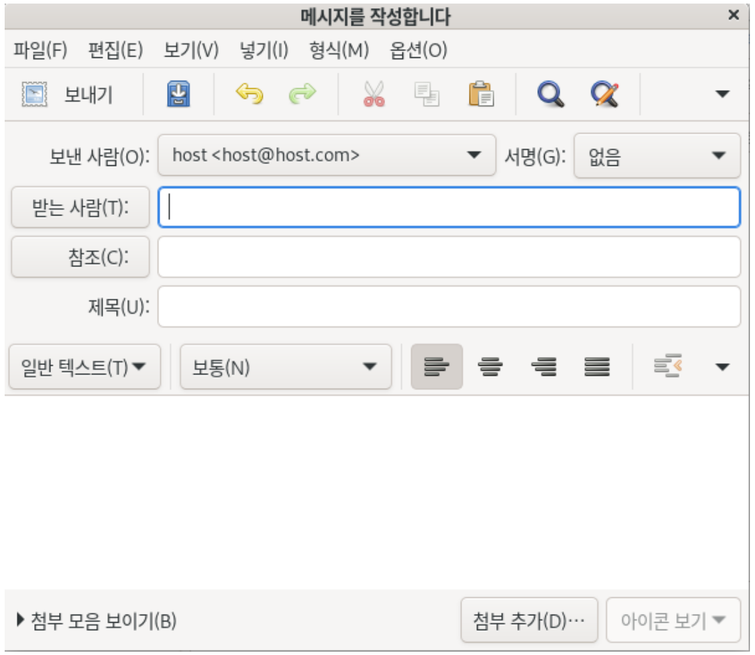
서버 구성

Mail

서버 동작 확인





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서버 구성

MariaDB

설치 및 확인

```
[root@localhost ~]# dnf -y install mariadb-server
[root@localhost ~]# rpm -qa mariadb-server
mariadb-server-10.5.29-3.el9_7.x86_64
```

패키지 설치

패키지 설치 확인

설치 완료

서비스 시작 및 활성화 확인

```
[root@localhost ~]# systemctl restart mariadb
[root@localhost ~]# systemctl enable mariadb
[root@localhost ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-01-15 17:17:32 KST; 6s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 10020 (mariadbd)
```

서비스 재시작

서비스 활성화

활성화 확인

방화벽 설정

```
[root@localhost ~]# firewall-cmd --permanent --add-port=3306/tcp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-port
3306/tcp 3389/tcp
```

방화벽 허용

방화벽 재가동

방화벽 가동 확인

05

서버 구성

MariaDB

DB 설정

관리자 비밀번호 설정

```
[root@localhost ~]# mysqladmin -u root password
```

```
MariaDB [(none)]> grant all on *.* to root@'%' identified by '1234';
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.000 sec)
```

외부 접속 허용 권한 부여

서버 동작 확인

```
[root@localhost ~]# mysql -h 192.168.111.100 -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
3 rows in set (0.000 sec)
```

SQL 원격 접속

THANK YOU

3조

한준영, 서승덕, 백우진, 양혜원

Rocky